Mon. Not. R. astr. Soc. (1990) **245,** 453–469

Reaction rates and energy distributions for elementary processes in relativistic pair plasmas

P. S. Coppi and R. D. Blandford
Theoretical Astrophysics 130-33, California Institute of Technology, Pasadena, California 91125, USA

Accepted 1990 March 9. Received 1990 January 5; in original form 1989 July 10

SUMMARY

Reaction rates are given for two-body processes in an isotropic-pair plasma. The rates are accurate over mildly relativistic energies and are given as functions of the incident particle energies rather than integrated over a thermal distribution. The processes considered are: Compton scattering, pair annihilation, two-photon pair production, Coulomb scattering, and e–e bremsstrahlung. For the first three processes, the mean energy of the final states together with the dispersion in energy about their mean energies are computed. Approximate expressions are presented where these have proven useful. The results have been tabulated for use in kinetic equation studies of pair plasmas.

1 INTRODUCTION

The discovery of rapidly variable X-ray and γ-ray emission from active galactic nuclei (AGN), γ-ray bursters, and the Galactic Centre has prompted investigation of 'electron–positron pair plasmas'. [For a general discussion of AGN and pair plasmas, see Svensson (1986), Zdziarski & Lightman (1987); for γ-ray bursters see Lamb (1988), Zdziarski (1987); for the Galactic Centre see Ramaty & Lingenfelter (1987).] Most of the initial attention focused on pair plasmas characterized by a single temperature (see Bisnovatyi-Kogan, Zel'dovich & Sunyaev 1971; Takahara 1980; Zdziarski 1985; Guilbert & Stepney 1985 and references contained therein). Unfortunately, such plasmas do not easily reproduce observed AGN spectral features and are unlikely to be produced because of the relatively long thermalization time-scales expected (e.g. Stepney 1983; Kusunose 1987).

More recent theoretical studies of pair plasmas therefore attempt to solve self-consistently for the particle distributions. Two methods have been followed: solving the kinetic equations for the electron and photon distribution functions (e.g. Guilbert 1981; Fabian *et al.* 1986; Ghisellini 1987a; Lightman & Zdziarski 1987; Svensson 1987), and direct Monte-Carlo simulations (e.g. Stern 1985; Novikov & Stern 1986). Both approaches are useful, the former when the computation must encompass a large range of energies or time-dependence is relevant, the latter when complex radiative transfer considerations are needed. The kinetic equation approaches employed to date, however, have been hampered by the use of oversimple approximations for interaction rates and scattered/emitted particle distributions. It is our purpose here to improve on some of these approximations.

In this paper, we re-examine the kinetic approach and set up a systematic framework for dealing with the 'primary' pair processes of Compton scattering, pair annihilation, and two-photon pair production. For these processes, we give both exact expressions and useful approximations for interaction rates, mean scattered energies, and dispersions about these mean energies as functions of the incident particle energies. The incident particle distributions are assumed isotropic and unpolarized. From these quantities we then construct an approximation for the energy distribution of the outgoing particles. The resulting formalism is designed to improve the computation of pair and high-energy photon yields. Also discussed are the processes of Coulomb scattering and e–e bremsstrahlung under approximations useful for kinetic calculations. The rate equations developed here are used in a time-dependent pair plasma code (Coppi 1990).

2 COMPTON SCATTERING

2.1 Kinematics

Compton scattering is most conveniently parametrized using two electron (or positron) rest frame quantities: the scattering angle θ between the ingoing and outgoing photon momenta, and the azimuthal angle ϕ. Assume now that the incoming electron and photon have (lab frame) energies $\gamma\,[=(1-\beta)^{-1/2}]\,m_e c^2$ and $\omega' m_e c^2$, respectively, and let the outgoing (scattered) photon have energy $\omega m_e c^2$. Using standard kinematic relations (e.g. Rybicki & Lightman 1979), we obtain

$$\omega=\frac{\gamma[\gamma(1-\beta\mu)+\gamma\beta\cos\theta(\mu-\beta)+\beta\sin\theta(1-\mu^2)^{1/2}\cos\phi]\,\omega'}{1+\gamma(1-\beta\mu)(1-\cos\theta)\,\omega'},\tag{2.1}$$

where μ is cosine of the (lab frame) angle between the incoming electron and photon-momentum vectors. The relevant scattering cross-section is given by the polarization-averaged Klein–Nishina cross-section, $\sigma_{\rm KN}$ (e.g. Berestetski, Lifshitz & Pitaevski 1982, p. 356, hereafter BLP). Ignoring induced scattering effects and assuming homogeneous, isotropic distributions, the photon occupation number $n(\omega)$ will then evolve according to

$$dn(\omega)/dt=-n(\omega)\int d\gamma N(\gamma)\,R(\omega,\gamma)+\iint d\omega' d\gamma P(\omega;\omega',\gamma)\,R(\omega',\gamma)\,n(\omega')\,N(\gamma).\tag{2.2}$$

Here $n(\omega)$ is the number of photons per unit volume with energies between ω and $\omega+d\omega$, $N(\gamma)$ is the analogous electron energy distribution, $R(\omega,\gamma)$ is the scattering rate between photons of energy ω and electrons of energy γ, and $P(\omega;\omega',\gamma)$ is the probability that the outgoing photon has energy ω given that the incoming photon and electron had energies ω' and γ, respectively. [$P(\omega;\omega',\gamma)$ is normalized so that $\int d\omega P(\omega;\omega',\gamma)=1$.]

2.2 Scattering rate

The angle-averaged scattering rate is given by

$$\begin{aligned}R(\omega,\gamma)&=c\int_{-1}^{+1}\frac{d\mu}{2}(1-\beta\mu)\,\sigma_{\rm KN}(\beta,\omega,\mu)\\&=\frac{3c\sigma_{\rm T}}{32\gamma^2\beta\omega^2}\int_{2\gamma(1-\beta)\omega}^{2\gamma(1+\beta)\omega}dx\left[\left(1-\frac{4}{x}-\frac{8}{x^2}\right)\ln(1+x)+\frac{1}{2}+\frac{8}{x}-\frac{1}{2(1+x)^2}\right],\end{aligned}\tag{2.3}$$

where $x=2\gamma(1-\beta\mu)\,\omega$. The integration over the variable x may be done analytically (e.g. Aharonian & Atoyan 1981a; Protheroe 1986), but the resulting expression which contains a dilogarithm is not very useful. $R(\omega,\gamma)$ is easily evaluated numerically.

Standard asymptotic forms for the rate R are:

(i) $\beta\ll 1$ and $\omega\ll 1$ (non-relativistic case),

$$R(\omega,\gamma)\approx c\sigma_{\rm T}(1-2\gamma\omega);\tag{2.4}$$

(ii) $\omega\ll 1/[2\gamma(1+\beta)]$ (i.e. $x\ll 1$),

$$R(\omega,\beta)\approx c\sigma_{\rm T}\left[1-\frac{2\gamma\omega}{3}(3+\beta^2)\right];$$

(iii) $\omega\gg 1/4\gamma$ and $\gamma\gg 1$ (extreme relativistic case),

$$R(\omega,\gamma)\approx\frac{3c\sigma_{\rm T}}{8\gamma\omega}\ln(4\gamma\omega).\tag{2.5}$$

In the 'mildly relativistic' scattering regime (e.g. $2\lesssim\gamma\lesssim 30$, $0.01\lesssim\omega\lesssim 30$) a useful interpolation for R is given by

$$R(\omega,\gamma)\approx R(\omega\gamma)=\frac{3c\sigma_{\rm T}}{8\gamma\omega}\left\{\left[1-\frac{2}{\gamma\omega}-\frac{2}{(\gamma\omega)^2}\right]\ln(1+2\gamma\omega)+\frac{1}{2}+\frac{4}{\gamma\omega}-\frac{1}{2(1+2\gamma\omega)^2}\right\}.\tag{2.6}$$

This expression is good to within 10 per cent for all energies and becomes exact in the limits $\beta\to 0$ and $\gamma\omega\to\infty$. A graph comparing the approximate to the exact values for $\gamma=20$ is given in Fig. 1. Note that in equation (2.6), there is only one independent variable, $(\gamma\omega)$. The step function (Klein–Nishina 'cut-off') approximation $R=\sigma_{\rm T}cH(1-\gamma\omega)$ (e.g. Fabian *et al.* 1986) does not work well in this scattering regime.

2.3 Scattered-photon distribution

The scattered photon energy distribution $P(\omega;\omega',\gamma)$ was derived by Jones (1968). (Unfortunately, the formulae given there appear to contain several misprints – see the Appendix.) The most common approximation to $P(\omega;\omega',\gamma)$ is the 'delta-function

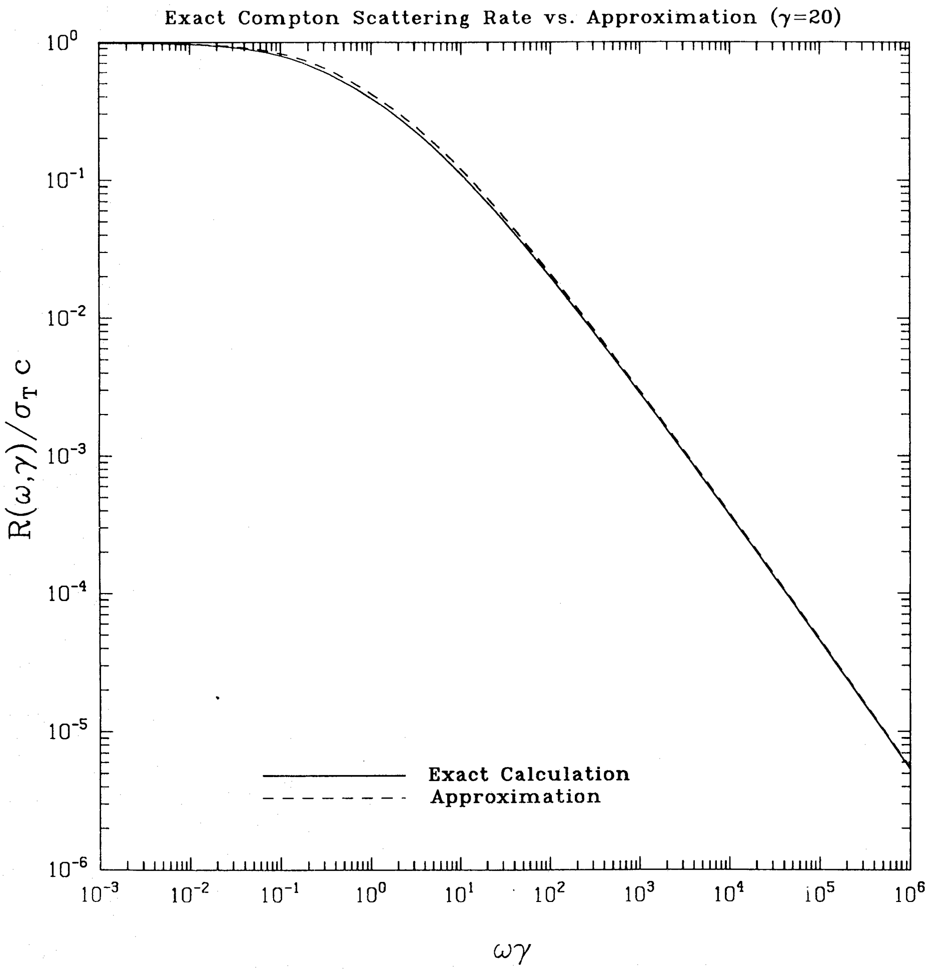

Figure 1. The Compton-scattering rate $R(\omega, \gamma)$ versus the approximation given in equation (2.6). The electron energy has been fixed at $\gamma = 20$. Note that $R(\omega, \gamma)$ begins to deviate significantly from $\sigma_T c$ well before $\omega\gamma = 1$.

approximation' $P(\omega; \omega', \gamma) \approx \delta(\omega - \langle\omega\rangle)$ with $\langle\omega\rangle$, the mean scattered-photon energy, approximated by $\frac{4}{3}\gamma^2\omega'$. To avoid negative scattered-electron energies and further simplify the equations, a cut-off approximation to the rate is also used, i.e. $R(\omega, \gamma) \approx \sigma_T c H(1 - \frac{3}{4}\omega\gamma)$. The inadequacy of this combination of approximations is shown in Fig. 2.

Use of the correct scattering rate and $\langle\omega\rangle$ significantly improves accuracy (e.g. Fig. 2), but reveals another limitation of delta-function approximations. The neglect of 'dispersion' in scattered-photon energies about $\langle\omega\rangle$ often results in equilibrium spectra which are too soft at the high-energy end (e.g. Fig. 2 again). This has serious consequences in calculations with significant two-photon pair production (e.g. Fabian *et al.* 1986; Lightman & Zdziarski 1987) as it leads to corresponding errors in the distribution of created pairs and propagates the errors in the high-energy photon distribution to lower energies. Errors of order unity at intermediate photon energies ($10^{-3} \lesssim \omega \lesssim 1$) can also be expected in this case.

To date, most attempts to include some form of dispersion have relied on a double expansion in ω'/γ and γ^2 (Jones 1968,equation 44; Blumenthal & Gould 1970; Aharonian & Atoyan 1981a; Aharonian, Kirillov-Ugryumov & Vardanian 1985; Zdziarski 1988). Unfortunately, the expansion fails when the scattered energy ω is not much greater than the incident energy ω' and gives quite misleading results when downscattering is important (e.g. Fig. 2). This approximation is also computationally intensive and not well-suited for calculations where many iterations are required.

We have experimented with several schemes to allow for dispersion in the scattered-photon distribution and found the best approximation to be

$$P'(\omega; \omega', \gamma) = \frac{1}{2D(\omega', \gamma)} H[D(\omega', \gamma) - |(\omega - \langle\omega\rangle)|], \tag{2.7}$$

where $H(x)$ is the Heaviside function, $D(\omega', \gamma) = \min\{[\sqrt{3\langle(\Delta\omega)^2\rangle}], (\langle\omega\rangle - \omega_{\min}), (\omega_{\max} - \langle\omega\rangle)\}$, and $\langle(\Delta\omega)^2\rangle$ is the mean square dispersion about the mean scattered energy. Here $\omega_{\min}$ and $\omega_{\max}$ are either the minimum and maximum photon energies handled by the numerical code under consideration, or 0 and ∞ in the general case. This procedure always gives the correct value for $\langle\omega\rangle$ and a good approximation for $\langle\omega^2\rangle$, the second moment of the scattered-photon distribution.

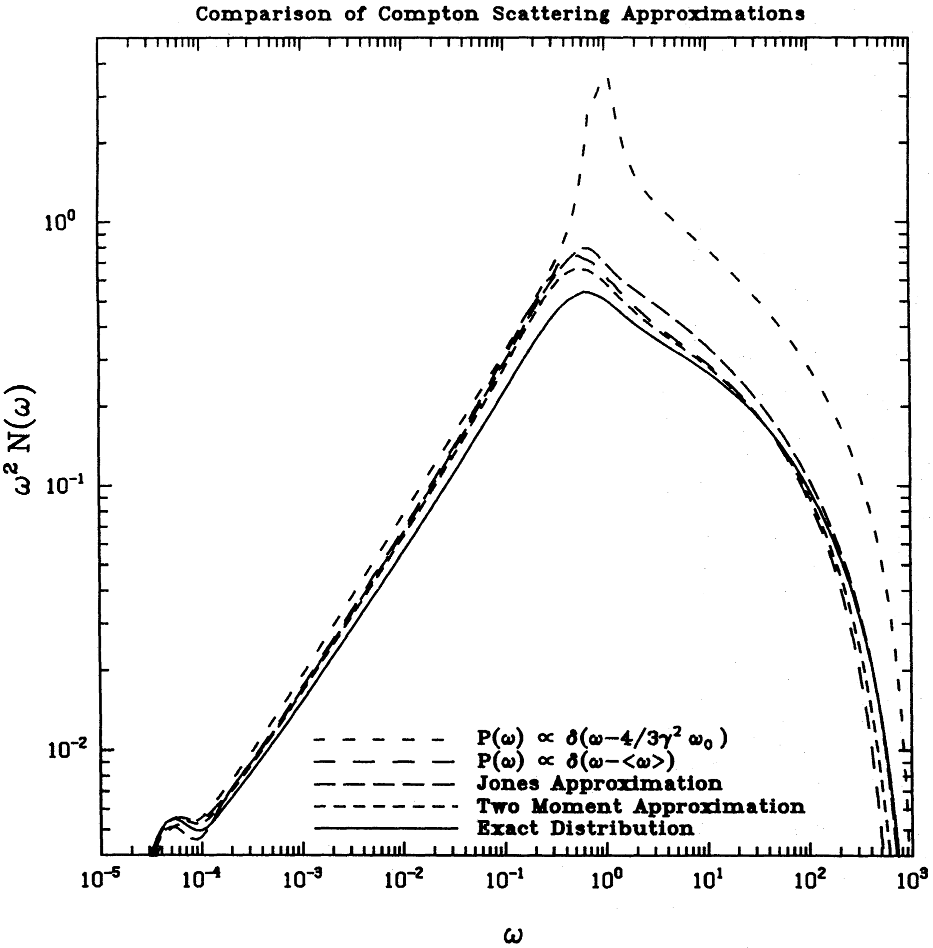

Figure 2. Comparison of the results of using various approximations to the distribution of Compton-scattered photons $P(\omega; \omega', \gamma)$. The electron distribution used was generated by running a 'photon-starved' model ($l_s/l_e = 0.03$) with steep ($\Gamma = 2.4$, $\gamma_{max} = 10^3$) power-law pair injection. [See, for example, Lightman & Zdziarski (1987); the model corresponds roughly to their model 'h'. l_s/l_e is the ratio of the injected soft photon luminosity to the injected-pair luminosity, and the pair-injection rate goes as $\gamma^{-\Gamma}$ up to a maximum injection energy γ_{max}.] For the calculation shown, the electron distribution was held fixed and the photon distribution that would be in kinetic equilibrium with it was then determined by adding up several 'orders' of Compton scattering (e.g. see Ghisellini 1987b). Note that in this calculation electrons with low γ are very important and that many (~ 10) orders of scattering contribute. The $P(\omega) \propto \delta(\omega - 4/3\gamma^2\omega_0)$ scattering approximation employs a Klein–Nishina cut-off for the scattering rate and is the same as that used in Lightman & Zdziarski (1987) and (modulo a factor of 4/3) Fabian *et al.* (1986). The 'corrected' [$P(\omega) \propto \delta(\omega - \langle \omega \rangle)$] delta-function approximation is the same except that the correct scattering rate and average scattered energy $\langle \omega \rangle$ are used. The 'Jones' approximation uses the approximate scattered distribution derived in Jones (1968) and employed, for example, in Zdziarski (1988). The 'two-moment' approximation is the one presented in this paper, and the 'exact' scattered distribution was computed using equation (A1.2). For $\omega < 100$, the maximum error of the 'corrected' delta-function approximation is $\approx +50$ per cent, while the two-moment approximation has a maximum error of $\approx +25$ per cent. The Jones approximation has a maximum deviation of ≈ 50 per cent at $\omega \sim 1$. Note the premature roll-off of both the corrected delta-function and two-moment approximations at the high-energy end of the photon spectrum.

The most serious limitations of this scheme arise from the 'top hat' shape assumed for $P'(\omega; \omega', \gamma)$. The real scattered distribution is often peaked and may be quite asymmetric. The electron and photon distributions typically considered in AGN models (extended power laws) appear broad enough to smooth away most of the effects of this discrepancy. However, as seen in Fig. 2, there may be a residual deficit of high-energy photons.

Based on several comparisons with results obtained using the exact probability distribution, errors up to ~ 25 per cent in the photon distribution are found for values of $\omega \lesssim 0.1\omega_{max}$ when pair production is not important. (ω_{max} is the maximum photon energy for the problem under consideration. When $\omega > \omega_{max}$, problems due to the inadequate dispersion discussed above set in, and the errors can be significantly larger.) For models with significant pair production, the maximum deviation seen was of the order of 40 per cent, with the typical error, however, still $\lesssim 25$ per cent (for $\omega \lesssim 0.1\omega_{max}$). Including dispersion even in this simplified manner produces significant gains in accuracy over a delta-function approximation, especially at the highest energies. The resulting scheme is quite fast and vectorizable.

In closing, we note that the approach described may be easily extended to thermal Comptonization problems (where the inclusion of dispersion is essential – omitting it leads to incorrect spectral indices and artificial features in the stationary spectra).

For temperatures $\tau=kT_e/m_e c^2 \lesssim 0.1$, using the top hat approximation with $\langle\omega\rangle$ and $\langle\omega^2\rangle$ averaged over a Maxwellian electron-energy distribution gives answers accurate to ~ 25 per cent, except at the highest energies where the spectrum again rolls off prematurely. For $\tau \gtrsim 0.1$, the fractional change in photon energy per scattering becomes large and this method fails. (A series of bumps appears in the expected power-law spectrum, corresponding to the distinct steps in the upscattering of a soft photon.) A better scattering kernel which remedies both this and the premature roll-off problem may be constructed by doing a Laguerre integration over top hat distributions, i.e. by using

$$K(\omega,\omega',\tau)=n_\mathrm{T}\left(\sum_{i=1}^{N} c_i P'(\omega,\omega',\gamma_i)\,R(\omega',\gamma_i)\right)\Bigg/\sum_{i=1}^{N} c_i, \tag{2.8}$$

where $c_i=\omega_i\gamma_i(\gamma_i^2-1)^{1/2}$, $\gamma_i=x_i\tau+1$, and x_i and ω_i are the zeros and weight factors for the Nth Laguerre polynomial (see Abramowitz & Stegun 1965, p. 923). Here n_T is the total number of thermal electrons, and $K(\omega,\omega',\tau)$ is the scattering rate of photons with energy ω' into energy ω by those electrons. For $N=6$, reasonable agreement ($\lesssim 30$ per cent) with Monte-Carlo calculations is obtained for temperatures up to $\tau\sim 10$. This kernel proved convenient in our numerical applications as it used the same tables we had computed to deal with non-thermal Comptonization. With regards to alternative kernels, we remark that a Gaussian kernel [with $\sigma=\langle(\Delta\omega)^2\rangle^{1/2}$, e.g. as used in Kusunose & Takahara 1985] works better than the simple top hat approach at low temperatures (it does not roll off quite as prematurely), but fails in a similar manner for $\tau\gtrsim 0.1$.

2.4 Mean scattered photon energy, $\langle\omega\rangle$

To derive the required quantity $\langle\omega\rangle$, it is much easier to work with the angular probability distribution $P_{\mu,\alpha}(\mu,\alpha;\omega',\gamma)$, where $\alpha=\cos(\theta)$ than with $P(\omega;\omega',\gamma)$. $\langle\omega\rangle$ is then given by

$$\langle\omega\rangle(\omega',\gamma)=\int_{-1}^{+1}\frac{d\mu}{2}\int_{-1}^{+1} d\alpha\,\bar{\omega}(\alpha,\mu)\,P_{\mu,\alpha}(\mu,\alpha;\omega',\gamma), \tag{2.9}$$

where

$$\bar{\omega}=\gamma\omega'[\gamma(1-\beta\mu)+\gamma\beta\alpha(\mu-\beta)]\left(\frac{x'}{x}\right),$$

and

$$P_{\mu,\alpha}(\mu,\alpha;\omega',\gamma)=c(1-\beta\mu)\frac{d\sigma(\mu,\alpha,\omega',\gamma)}{d\alpha}\Bigg/R(\omega',\gamma), \tag{2.10}$$

with

$$\frac{d\sigma(\mu,\alpha,\omega',\gamma)}{d\alpha}=\frac{3}{8}\sigma_\mathrm{T}\left(\frac{x'}{x}\right)^2\left(\frac{x}{x'}+\frac{x'}{x}-1+\alpha^2\right). \tag{2.11}$$

$x=\gamma\omega'(1-\beta\mu)$ and $x'=x/[1+(1-\alpha)x]$ are, respectively, the incident- and scattered-photon energies as viewed in the electron rest frame.

In evaluating this expression, it proves most convenient to perform the first integral analytically and the second numerically. (The second integration may be done analytically, however – see the expression for $\langle d\gamma/dt\rangle$ in Jones 1965.) Making the variable change (suggested to us by P. Madau) $\alpha\to x'$, the integrand becomes an easily integrated polynomial in x'. The end result is

$$\begin{aligned}\langle\omega\rangle(\mu,\omega',\gamma)=\frac{3\sigma_\mathrm{T}cx}{8\gamma\omega' R(\omega',\gamma)}\Bigg\{&\frac{2}{(1+2x)}[a+\gamma(x-2)-(2\gamma+a)/x-a(6x^{-2}+3x^{-3})]\\&+\frac{\ln(1+2x)}{x^2}[\gamma-a+3a(x^{-1}+x^{-2})]\\&+\frac{1}{2x}\left[1-\frac{1}{(1+2x)^2}\right][a(3x^{-2}+x^{-3})+(a+\gamma)x^{-1}+2\gamma]+\frac{1}{3}\left[1-\frac{1}{(1+2x)^3}\right]\\&\times[\gamma+a(x^{-1}+x^{-2})]-2ax^{-3}\Bigg\},\end{aligned} \tag{2.12}$$

where $a=\gamma^2\beta\omega'(\mu-\beta)$. The desired $\langle\omega\rangle$ is then

$$\int_{-1}^{+1}\frac{d\mu}{2}\langle\omega\rangle(\mu,\omega',\gamma).$$

(For scattering off a thermal-pair distribution, Barbosa (1982) has derived a somewhat complicated series expansion for $\langle\omega\rangle$ and $\langle\omega^2\rangle$.)

Some useful limiting forms for $\langle\omega\rangle$ are

$$\langle\omega\rangle\approx(1+\tfrac{4}{3}\beta^2-\omega')\,\omega',\quad \beta,\omega'\ll 1,\tag{2.13}$$

and

$$\langle\omega\rangle\approx\tfrac{4}{3}\gamma^2\omega',\quad 1\ll\gamma\ll\omega'^{-1}.\tag{2.14}$$

Note the conditions for the validity of this asymptotic form. In particular, when there are significant numbers of scatterings with $\omega'\gamma\sim 1$ (e.g. Fig. 2), it may be a very bad approximation to use for $\langle\omega\rangle$.

These limits unfortunately do not suggest any approximations valid over a sufficiently wide range of γ and ω'. Also, because $\langle\omega\rangle$ is a function of two variables, other standard approximation techniques (e.g. fitting with a polynomial) do not appear very useful in providing an analytic approximation both simple and accurate enough for use in a calculation. A good, relatively fast numerical approximation may always be constructed through the use of an interpolation table. A table of 20×20 entries covering the range $1<\gamma<10^2$, $10^{-2}<\omega'<10^2$ gives values accurate to ~5 per cent.

2.5 Dispersion about $\langle\omega\rangle$

Following the previous section, we write

$$\langle\omega^2\rangle(\omega',\gamma)=\int_{-2}^{+1}\frac{d\mu}{2}\int_{-1}^{+1}d\alpha\bar{\omega}^2P_{\mu,\alpha}(\mu,\alpha;\omega',\gamma)\tag{2.15}$$

where

$$\bar{\omega}^2=\frac{\gamma^2\omega'^2\{[\gamma(1-\beta\mu)+\gamma\beta\alpha(\mu-\beta)]^2+\frac{1}{2}\beta^2(1-\alpha^2)(1-\mu^2)\}}{[1+\gamma(1-\beta\mu)(1-\alpha)\,\omega']^2}.\tag{2.16}$$

The dispersion is then given by $\langle(\Delta\omega)^2\rangle=\langle\omega^2\rangle-\langle\omega\rangle^2$. Note that, as above, the integral over α may be done analytically. The result is too cumbersome to include here, however. $\langle\omega^2\rangle$ has been evaluated numerically and tabulated. We have found only two useful limiting forms. These are

$$\langle\omega^2\rangle\approx\frac{14}{5}\gamma^4\omega'^2\left(1-\frac{176}{35}\gamma\omega'\right),\quad 1\ll\gamma\ll\omega'^{-1}\tag{2.17}$$

and

$$\begin{aligned}R(\omega',\gamma)\langle\omega^2\rangle\approx{}&\frac{1}{64\beta\gamma^2\omega'^2}\{[6\gamma\omega'(1+\beta)(2\gamma^2+1)+6\gamma^2+3]\ln[2\gamma\omega'(1+\beta)+1]\\&-[6\gamma\omega'(1-\beta)(2\gamma^2+1)+6\gamma^2+3]\ln[2\gamma\omega'(1-\beta)+1]\},\\&+\frac{9}{32}(1-\beta^2)^2\gamma^3\omega'-\frac{58\gamma^2+1}{64\gamma\omega'}+\frac{7}{32}[2\gamma^2(1-\beta^2)+1].\qquad \omega'\gamma\gg1\end{aligned}\tag{2.18}$$

Using equation (2.5) for $R(\omega',\gamma)$, one obtains an approximation for $\langle\omega^2\rangle$ good to within 10 per cent by $\gamma\omega'\sim40$.

3 PAIR ANNIHILATION

3.1 Cross-section

As in past work (e.g. Aharonian & Atoyan 1981b; Ramaty & Mészáros 1981; Svensson 1982; Aharonian, Kirillov-Ugryumov & Kotov 1983b), we assume that the incident-electron and positron distributions are isotropic and unpolarized. The invariant differential cross-section (e.g. BLP, section 88.15) may be integrated to give the total cross-section as a function of the centre-of-mass speed β' of the incident particles:

$$\sigma(\beta')=\frac{1}{2}\int d\sigma=\frac{3\sigma_T(1-\beta'^2)}{32\beta'}\left[\frac{3-\beta'^4}{\beta'}\ln\left(\frac{1+\beta'}{1-\beta'}\right)-2(2-\beta'^2)\right]. \tag{3.1}$$

Note that to avoid double counting in the derivation of the *total* cross-section, the cross-section obtained by integrating $d\sigma$ must be divided by two since the two photons produced are *identical* particles.

3.2 Pair annihilation rate

The pair annihilation rate is

$$R(\gamma_-,\gamma_+)=c\int_{-1}^{+1}\frac{d\mu}{2}f_{\rm kin}(\beta_-,\beta_+,\mu)\,\sigma(\beta_-,\beta_+,\mu), \tag{3.2}$$

where

$$f_{\rm kin}(\beta_-,\beta_+,\mu)=\sqrt{\beta_-^2+\beta_+^2-\beta_-^2\beta_+^2(1-\mu^2)-2\beta_-\beta_+\mu}, \tag{3.3}$$

and $\sigma(\beta_-,\beta_+,\mu)$ is the total cross-section (equation 3.1) with

$$\beta'^2=\frac{[\gamma_+\gamma_-(1-\mu\beta_+\beta_-)-1]}{[\gamma_+\gamma_-(1-\mu\beta_+\beta_-)+1]}.$$

The integral can be performed analytically, e.g. Svensson (1982, equation 18).

Useful asymptotic forms for $R(\gamma_-,\gamma_+)$ are

(i) $\beta_-,\beta_+\ll 1$ (non-relativistic regime),

$$R(\gamma_-,\gamma_+)\to\tfrac{3}{8}c\sigma_T; \tag{3.4}$$

(ii) β_- or $\beta_+\ll 1$ (useful for energetic pairs impinging on a non-relativistic pair population),

$$R(\gamma_-,\gamma_+)\to R(\beta)=c\beta\sigma(\beta'), \tag{3.5}$$

where β represents the non-vanishing β_- or β_+, $\sigma(\beta')$ is defined above, and β' reduces to $(\gamma-1/\gamma+1)^{1/2}$ with $\gamma=(1-\beta^2)^{-1/2}$.

(iii) $\gamma_-\gamma_+\gg 1$ (relativistic regime),

$$R(\gamma_-,\gamma_+)\to R(\gamma_-\gamma_+)=\tfrac{3}{8}c\sigma_T\ln(\gamma_-\gamma_+)/\gamma_-\gamma_+. \tag{3.6}$$

One should note that for case (iii), R converges slowly to the limit given. The difference between the two only decreases from 10 to 4 per cent as $\gamma_-\gamma_+$ increases from 10 to 10^8.

A simple approximation to the exact $R(\gamma_-,\gamma_+)$ may be found by interpolating between the limiting cases (i) and (iii). The expression we have found to work best is

$$R(\gamma_-,\gamma_+)\approx R(x)=\frac{3\sigma_T c}{8x}[x^{-1/2}+\ln(x)],\quad x=\gamma_-\gamma_+. \tag{3.7}$$

This expression is accurate to within 14 per cent for all γ_-,γ_+. A graph showing the exact and approximate pair annihilation rate for $\gamma_-=5$ is shown in Fig. 3.

3.3 Average photon energy, $\langle\omega\rangle$, and dispersion about $\langle\omega\rangle$

Energy conservation and the fact that the created photons are identical particles implies

$$\langle\omega\rangle=\frac{\gamma_-+\gamma_+}{2}. \tag{3.8}$$

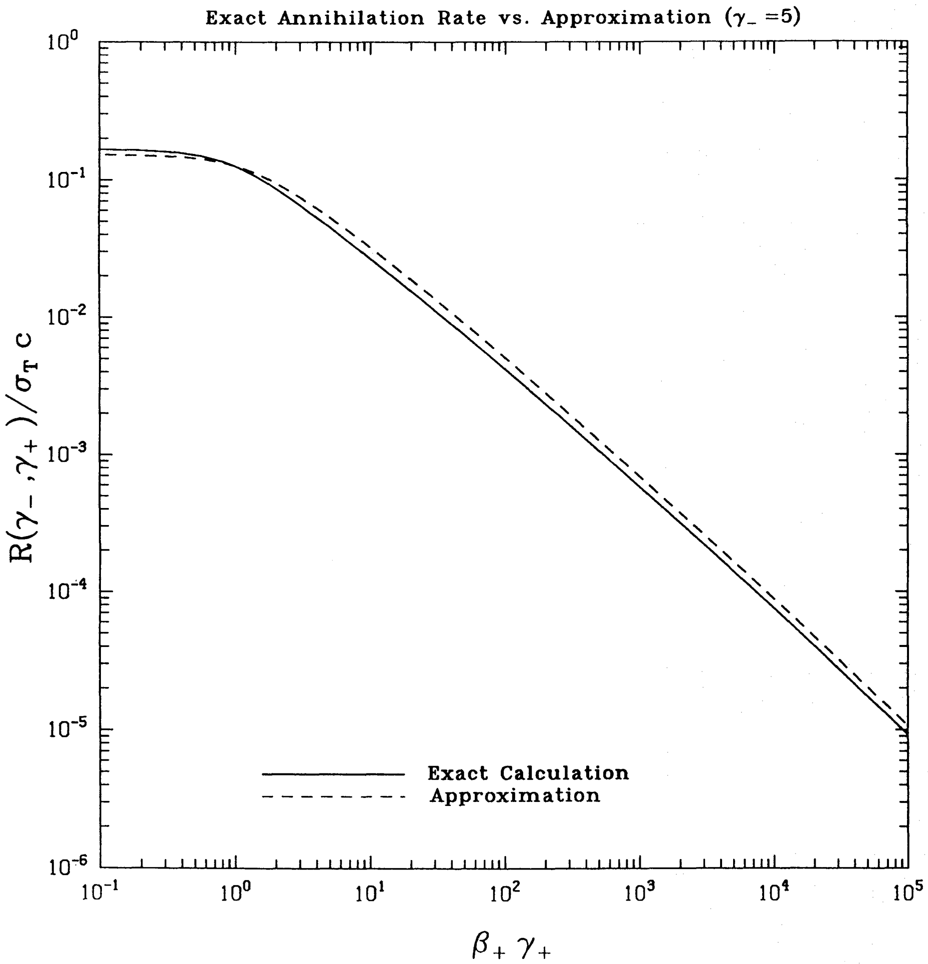

Figure 3. The exact annihilation rate $R(\gamma_-, \gamma_+)$ versus the approximation given in equation (3.7). The electron energy has been fixed at $\gamma_- = 5$. The maximum deviation between the two rates is less than 14 per cent.

Following the example of Compton scattering,

$$\langle \omega^2 \rangle = \frac{c}{2R(\gamma_-, \gamma_+)} \int_{-1}^{+1} \frac{d\mu}{2} f_{\rm kin}(\mu) \int_{-1}^{+1} d\alpha \left[\frac{d\sigma(\alpha, \mu)}{d\alpha} \right] [\bar{\omega}^2(\alpha, \mu)], \tag{3.9}$$

where α is the cosine of the centre-of-mass scattering angle, and

$$\frac{d\sigma}{d\alpha} = \frac{3\sigma_{\rm T}}{16\beta' \gamma'^2} \left[\frac{1 + \beta'^2(2 - \alpha^2)}{(1 - \beta'^2 \alpha^2)} - \frac{2\beta'^4 (1 - \alpha^2)^2}{(1 - \beta'^2 \alpha^2)^2} \right], \tag{3.10}$$

$$\bar{\omega}^2(\alpha, \mu) = \frac{\gamma'^2}{(1 - \beta_{\rm CM}^2)} \left\{ 1 + 2\alpha\beta_{\rm CM} \cos \xi' + \beta_{\rm CM}^2 \left[\frac{1 - \alpha^2}{2} + \frac{1}{2} \cos^2 \xi' (3\alpha^2 - 1) \right] \right\}, \tag{3.11}$$

$$\beta_{\rm CM} = \frac{(\gamma_-^2 \beta_-^2 + \gamma_+^2 \beta_+^2 + 2\gamma_+ \gamma_- \beta_+ \beta_- \mu)^{1/2}}{\gamma_+ + \gamma_-}, \tag{3.12}$$

$$\cos \xi' = \beta_{\rm CM}^{-1} \beta'^{-1} \left(\frac{\gamma_+ - \gamma_-}{\gamma_+ + \gamma_-} \right), \tag{3.13}$$

and $\gamma' = (1 - \beta'^2)^{-1/2}$. There do not appear to be any useful analytic approximations for $\langle \omega^2 \rangle$. ($\langle \omega^2 \rangle$ was calculated numerically and tabulated.)

3.4 Comparison with exact results

To demonstrate the accuracy of the prescription just presented, we calculate (numerically) the photon distribution produced by power-law 'non-thermal' pairs annihilating on cold pairs. This corresponds to the 'cosmic-ray' case discussed in Aharonian *et al.* (1983b) and Svensson (1982) for which a relatively simple analytic expression for the annihilation spectrum exists (e.g. Svensson 1982, equation 41). This is also the typical situation encountered in codes of the type described in Paper II (Coppi 1990), where cooling (approximately power-law) non-thermal pairs annihilate primarily with cool ($kT \lesssim 0.01 m_e c^2$) thermal pairs. As can be seen in Fig. 4, for example, the approximation is adequate, with an error of less than 10 per cent over the range $0.8 < \omega < 35$, the range in which most of the annihilation photons lie. Note, though, that this approximation tends to overestimate the production rate of low-energy photons. In applications of the type discussed in Paper II, however, these photons may usually be ignored as the photon distribution at such energies is dominated by photons from other sources.

4 PHOTON–PHOTON PAIR PRODUCTION

4.1 Cross-section

As in previous work (e.g. Gould & Schréder 1967; Aharonian, Atoyan & Nagapetyan 1983a; Aharonian *et al.* 1985; Zdziarski 1988), we assume that the distribution of incident photons is unpolarized and isotropic. Photon–photon pair production is the inverse process to pair annihilation. The cross-section is therefore identical to equation (3.1) except for a different phase-space factor and a factor of 2 due to the fact the electrons and positrons produced are no longer identical particles, that is (e.g. BLP, section 89):

$$d\sigma_{\gamma\gamma\to ee} = \beta'^2 d\sigma_{ee\to\gamma\gamma}; \quad \sigma_{\gamma\gamma\to ee} = 2\beta'^2 \sigma_{ee\to\gamma\gamma}. \tag{4.1}$$

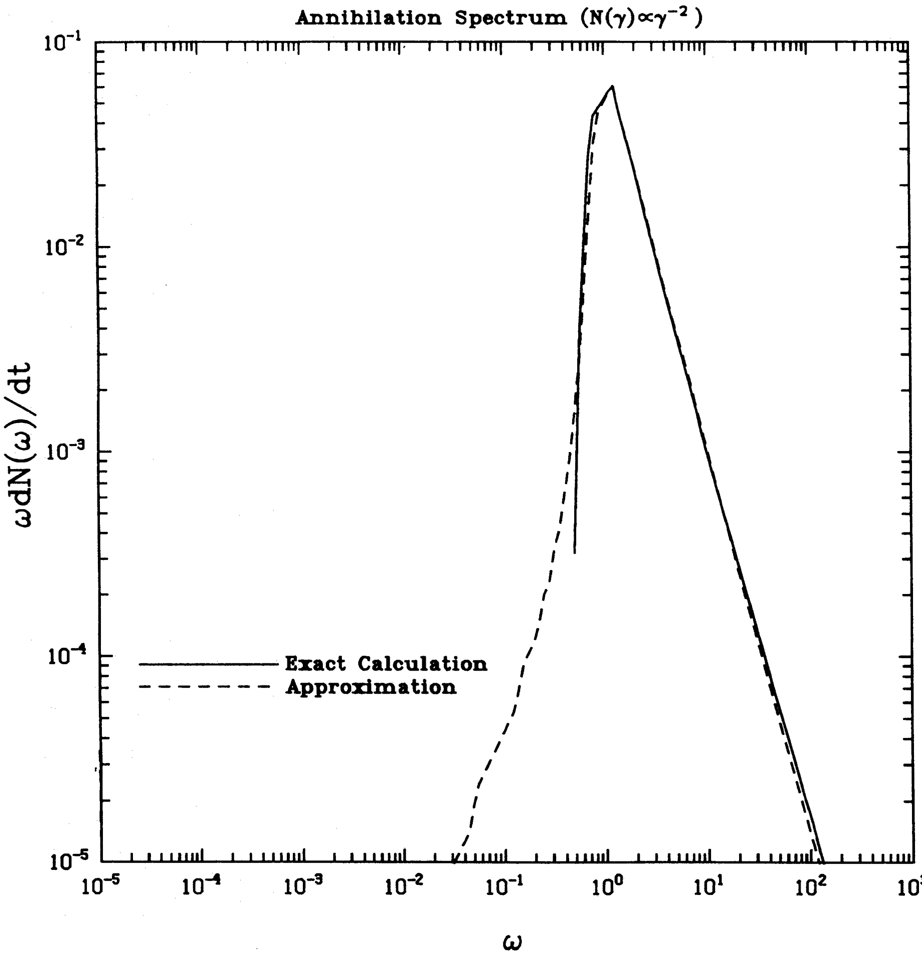

Figure 4. The annihilation spectrum (in arbitrary units) produced by power-law pairs [$N(\gamma) \propto \gamma^{-2}$, $1.1 < \gamma < 10^3$] annihilating on cold ($\beta = 0$) pairs. The exact spectrum was calculated using equation (41) of Svensson (1982) while the approximate spectrum was calculated using the scattering approximation presented in Section 3 of the text. For $0.8 \lesssim \omega \lesssim 35$, the maximum deviation between the two is less than 10 per cent, and less than 30 per cent for $0.75 \lesssim \omega \lesssim 500$.

The total cross-section as a function of β', the centre-of-mass speed of the electron and positron, is thus

$$\sigma(\beta')=\int d\sigma=\frac{3\sigma_{\mathrm{T}}(1-\beta'^2)}{16}\left[(3-\beta'^4)\ln\left(\frac{1+\beta'}{1-\beta'}\right)-2\beta'(2-\beta'^2)\right]. \tag{4.2}$$

4.2 Pair production (photon-annihilation) rate

The desired annihilation rate of photons into electron–positron pairs is given by

$$R(\omega_1,\omega_2)=c\int_{-1}^{\mu_{\max}}\frac{d\mu}{2}(1-\mu)\,\sigma(\omega_1,\omega_2;\mu), \tag{4.3}$$

where $\sigma(\omega_1,\omega_2;\mu)$ is the total cross-section (equation 4.2) with $\beta'=[1-2/\omega_1\omega_2(1-\mu)]^{1/2}$. The upper limit of integration $\mu_{\max}=\max(-1,1-2/\omega_1\omega_2)$ is determined by the requirement that β' remain real. Note that $R(\omega_1,\omega_2)$ is really a function of one variable, $x=(\omega_1\omega_2)$. A graph of this function is shown in Fig. 5 (see also Gould & Schréder 1967).

Two useful asymptotic limits for $R(\omega_1,\omega_2)=R(x)$ are

(i) $x\rightarrow 1$ (threshold),

$$R\rightarrow\tfrac{1}{2}c\sigma_{\mathrm{T}}(x-1)^{3/2}, \tag{4.4}$$

(ii) $x\gg 1$ (ultra-relativistic limit),

$$R\rightarrow\tfrac{3}{4}c\sigma_{\mathrm{T}}\ln(x)/x. \tag{4.5}$$

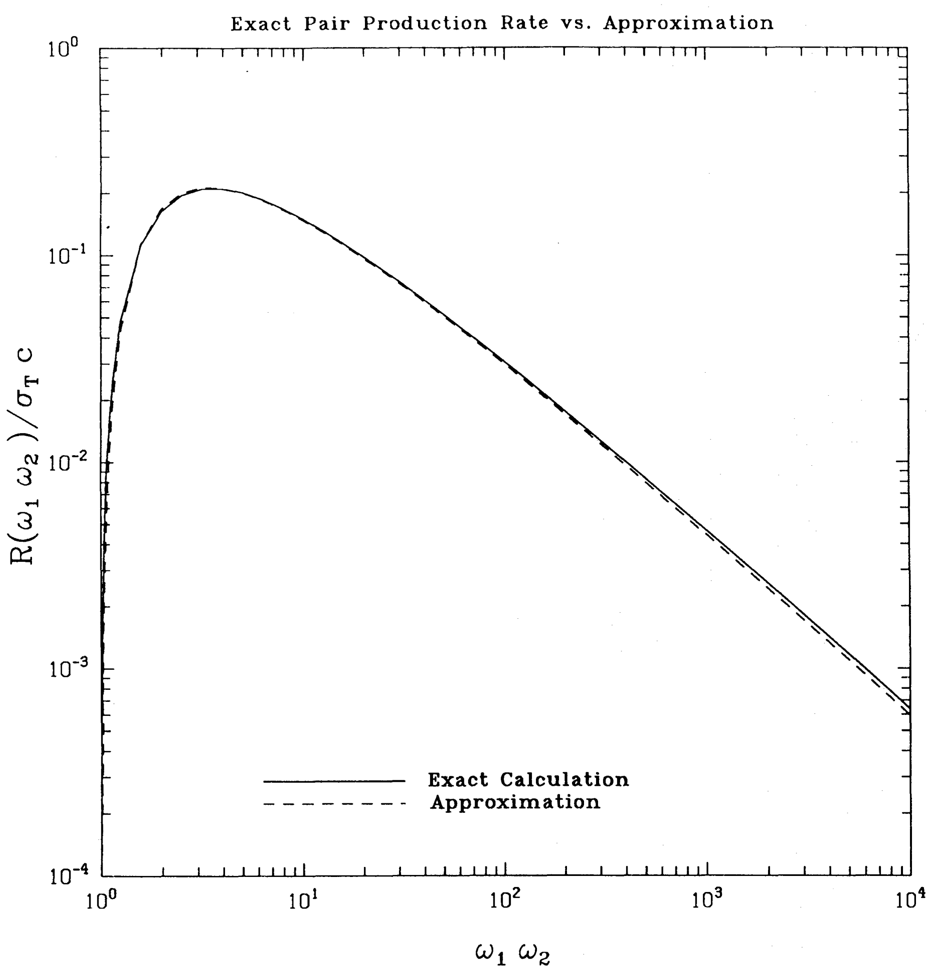

Figure 5. The pair-production rate $R(\omega_1\omega_2)$ versus the approximation given in equation (4.7). Over the range $1.3\lesssim\omega_1\omega_2\lesssim 10^4$, the maximum deviation between the two is less than 7 per cent.

As in the case of pair annihilation, R approaches the ultra-relativistic limit slowly.

A fairly good approximation may again be constructed by interpolating between the limiting cases, e.g.

$$R(x) \approx c\sigma_T \frac{(x-1)^{3/2}}{x^{5/2}} \left(\tfrac{1}{2}x^{-1/2} + \tfrac{3}{4}\ln x\right) H(x-1), \tag{4.6}$$

which has a maximum error of 12 per cent. Aharonian *et al.* (1983a) have derived a more complex alternative expression (equation 6 of their paper) with the same limiting behaviour, but accurate to better than 5 per cent at all energies. A simpler alternative to either, very accurate near the peak of $R(x)$ (at $x_{\text{peak}} \approx 3.7$), is given by

$$R(x) \approx 0.652 c\sigma_T \frac{(x^2-1)}{x^3} \ln(x)\, H(x-1). \tag{4.7}$$

Over the range $1.3 < x < 10^4$ (which usually dominates in calculations), this expression is accurate to better than 7 per cent (see Fig. 5).

4.3 Average pair energy, $\langle \gamma \rangle$, and dispersion about $\langle \gamma \rangle$

From energy conservation and symmetry considerations,

$$\langle \gamma \rangle = \frac{\omega_1 + \omega_2}{2}, \tag{4.8}$$

while $\langle \gamma^2 \rangle$ is given by

$$\langle \gamma^2 \rangle = \frac{c}{R(\omega_1, \omega_2)} \int_{-1}^{+1} \frac{d\mu}{2} (1-\mu) \int_{-1}^{+1} d\alpha \left[\frac{d\sigma(\alpha,\mu)}{d\alpha}\right] [\bar{\gamma}^2(\alpha,\mu)] \tag{4.9}$$

where α is the cosine of the centre-of-mass angle, and

$$\frac{d\sigma}{d\alpha} = \frac{3\sigma_T(\omega'^2-1)^{1/2}}{16\omega'^3} \left\{\frac{\omega'^2 + (\omega'^2-1)(2-\alpha^2)}{\omega'^2 - (\omega'^2-1)\alpha^2} - \frac{2(\omega'^2-1)^2(1-\alpha^2)^2}{[\omega'^2-(\omega'^2-1)\alpha^2]^2}\right\} \tag{4.10}$$

$$\bar{\gamma}^2(\alpha,\mu) = \frac{\omega'^2}{(1-\beta_{\text{CM}}^2)} \left\{1 + 2\alpha\beta'\beta_{\text{CM}}\cos\xi' + \beta_{\text{CM}}^2\beta'^2 \left[\frac{1-\alpha^2}{2} + \frac{1}{2}\cos^2\xi'(3\alpha^2-1)\right]\right\}, \tag{4.11}$$

$$\beta_{\text{cm}} = \frac{(\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2\mu)^{1/2}}{\omega_1 + \omega_2}, \tag{4.12}$$

$$\cos\xi' = (\omega_1 - \omega_2)(\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2\mu)^{-1/2}, \tag{4.13}$$

and $\omega' = \gamma' = (1-\beta'^2)^{-1/2}$. The dispersion was evaluated numerically.

4.4 Photon absorption probability

We comment briefly on an approximation that has been used to estimate the value of $d\tau_{\gamma\gamma}/ds(\omega)$, the absorption probability per unit path-length of a photon due to pair production. In calculations of the type carried out in Svensson (1987) and Lightman & Zdziarski (1987), the process of pair production is often quite significant in determining the overall shape of the equilibrium particle distributions. Accurately estimating the pair-injection spectrum produced by an annihilating-photon spectrum is thus important and requires a correspondingly accurate estimate of the absorption probability. [The absorption probability $d\tau/ds(\omega)$ is closely related to the pair-production rate $R(x)$, and for a general photon distribution $n(\omega)$, is given by $1/c \int_0^\infty dx\, R(x\omega)\, n(x)$.]

Svensson (1987, Appendix B.2) has shown that for a photon power law extending from $\omega = 0$ to $\omega = \infty$, $d\tau_{\gamma\gamma}/ds$ is given by the formula

$$d\tau_{\gamma\gamma}/ds(\omega) = \eta(\alpha)\,\sigma_T \frac{n(1/\omega)}{\omega}. \tag{4.14}$$

(Here α is the index of the power law, and η runs between 0.24 for $\alpha=0.5$ and 0.12 for $\alpha=1.0$.) A simple approximation can thus be constructed by taking this result and fixing the value of η at some intermediate value (usually 0.2). (One might arrive at the same estimate for $d\tau/ds$ by noting that $R(\omega_1, \omega_2)$ has a maximum value $\approx 0.2\sigma_T c$ for $\omega_2 \sim 1/\omega_1$. Hence,

$$d\tau/ds(\omega_1) \sim 0.2\sigma_T n\left(\frac{1}{\omega_1}\right)\left(\frac{1}{\omega_1}\right), \tag{4.15}$$

where the range of energies over which the integrand is taken to be 'large' is $\sim 1/\omega_1$.) However, as seen in Fig. 6, this approximation can lead to serious errors of order a factor 2 or more when applied to a realistic photon distribution.

4.5 Accuracy of present approximation

To check the validity of the dispersion prescription presented, we have compared the distribution of produced pairs calculated with our method against that calculated using the exact distribution (see Appendix A.2). A sample calculation is shown in Fig. 7. The results agree to better than 15 per cent at all energies, with the largest errors at the minimum and maximum pair energies. The good agreement reflects the fact that the exact distribution is always symmetric about $\langle\gamma\rangle$ and has a shape often resembling the 'top hat' shape assumed for the approximation.

Also shown in Fig. 7 are calculations based on the approximate distribution presented in equation (11) of Aharonian *et al.* (1985) and the delta-function approximation (where all pairs are produced at the average energy $\langle\gamma\rangle$). The Aharonian approximation is similar to the one made in Jones (1968) for Compton scattering and is valid in the regime $\omega_1 \gg \max(\omega_2, 1)$ (where ω_1 is the larger of the incoming photon energies). Use of the approximation works extremely well at high energies but

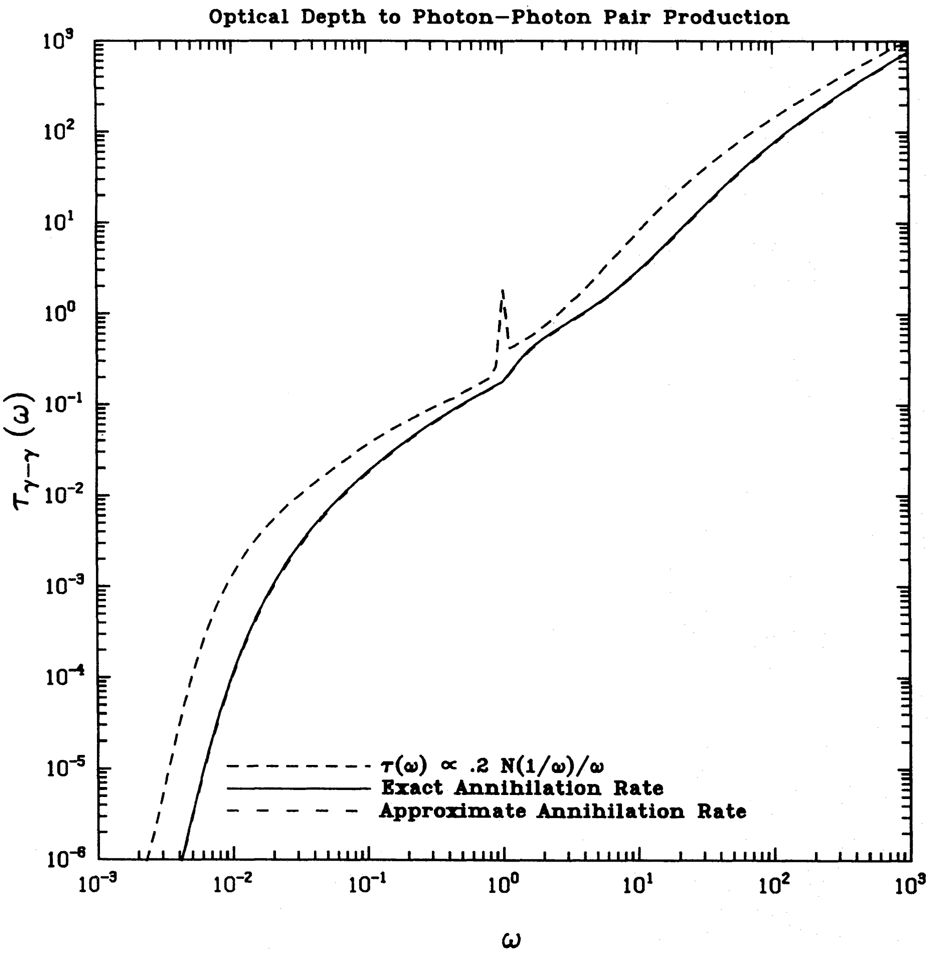

Figure 6. The 'optical depth' to photon–photon pair production (see Section 4.4) calculated for a photon distribution corresponding roughly to that of model 'P' in Lightman & Zdziarski (1987). The approximation $\tau_{\gamma\gamma}(\omega) \propto 2n(1/\omega)/\omega$ is the one employed in Svensson (1987) and Lightman & Zdziarski (1987). The other curves were calculated by integrating the photon distribution over the exact pair-production rate $R(\omega_1\omega_2)$ and the approximation given in equation (4.7). Using equation (4.7) gives an answer accurate to better than 2 per cent over the frequency range shown.

leads to errors of order unity at energies $\gamma \sim 1$. (There most of the pairs come from the annihilation of photons with $\omega_1 \sim \omega_2 \sim 1$.) As expected, use of the delta-function approximation also leads to errors of order unity, particularly at the highest and lowest pair energies. These disagreements can impact significantly the equilibrium pair distribution (and hence the equilibrium photon distribution) calculated by a code such as that of Lightman & Zdziarski (1987).

5 COULOMB SCATTERING

Coulomb scattering usually represents only a minor contribution to the overall energy balance, although substantial changes in the equilibrium pair and photon distributions can ensue if it is neglected. For many applications, the effects of Coulomb scattering may be treated as an energy exchange term between non-thermal and thermal pairs in the pair kinetic equations. (For a Fokker–Planck treatment, see Dermer & Liang 1989.) For the non-thermal pairs, this term takes the form

$$\dot{\gamma} = c \int_1^\infty d\gamma_1 P(\gamma_1) \int_{-1}^{+1} \frac{d\mu}{2} f_{\rm kin}(\gamma, \gamma_1, \mu) < \sigma \Delta \gamma >_\alpha (\gamma, \gamma_1, \mu) \tag{5.1}$$

(see Stepney 1983; Baring 1987a, b; Haug 1988). Here, $\dot{\gamma}$ is the rate of change of energy for an electron or positron of initial energy γ scattering off an isotropic thermal distribution of electrons or positrons described by

$$P(\gamma_1)\, d\gamma_1 = N_{\rm T} \frac{\gamma_1^2 \beta_1 \exp(-\gamma_1/\tau)}{\tau K_2(1/\tau)}\, d\gamma_1, \tag{5.2}$$

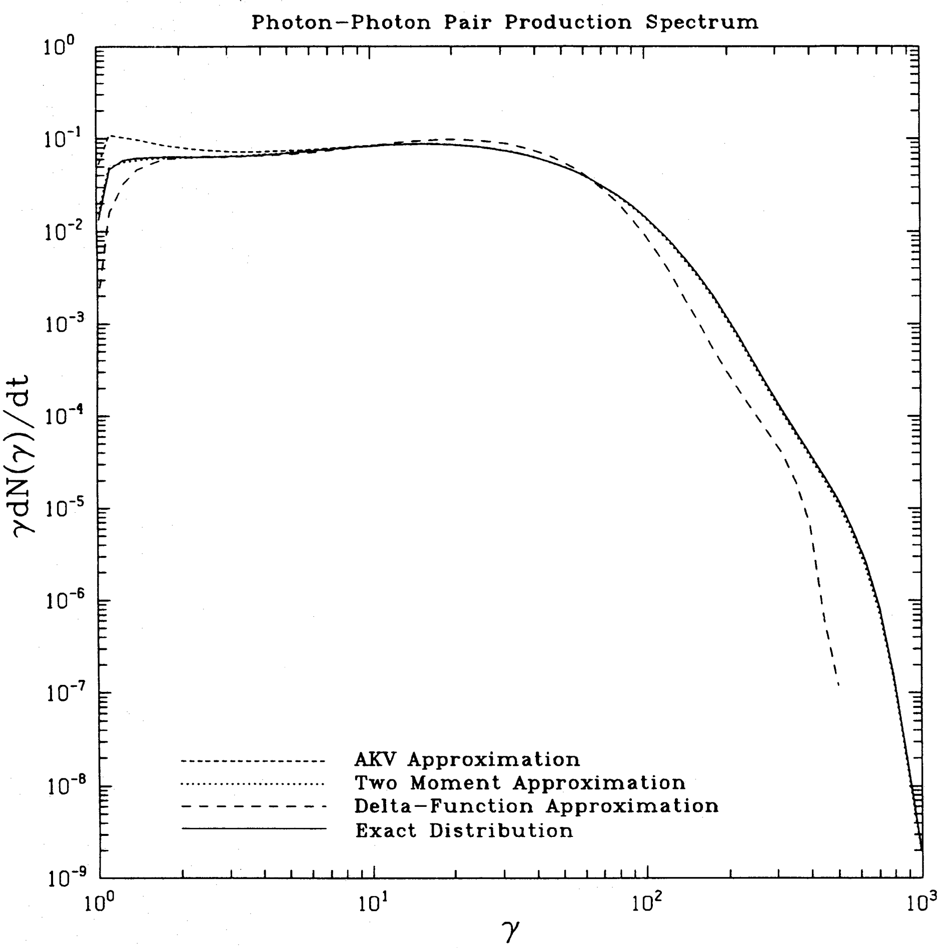

Figure 7. The pair 'spectrum' (in arbitrary units) produced by the annihilation of photons in the same distribution as considered for Fig. 6. The 'AKV' approximation uses the approximate distribution presented in Aharonian *et al.* (1985). Note that this approximation is not very good for $\gamma \sim 1$. The 'delta-function' approximation curve is calculated using the exact pair-production rate but assuming that the created pairs all have energy $\langle \gamma \rangle = (\omega_1 + \omega_2)/2$. The 'two-moment' approximation is the one presented in Section 4, and the 'exact' distribution of pairs was computed using equation (A2.1). The 'two-moment' and 'exact' results agree to within 15 per cent.

where $\tau = k_B T/m_e c^2$, $K_2(x)$ is a modified Bessel function, and N_T is the total number of thermal electrons or positrons. $f_{kin}(\gamma, \gamma_1, \mu)$ is the kinetic factor of equation (3.3), and

$$\langle \sigma\Delta\gamma\rangle_\alpha(\gamma, \gamma_1, \mu) = \int_{\alpha_{min}}^{\alpha_{max}} d\alpha \left[\frac{d\sigma(\alpha, \gamma, \gamma_1, \mu)}{d\alpha}\right] [\Delta\gamma(\alpha, \gamma, \gamma_1)], \tag{5.3}$$

where α is the cosine of the centre-of-mass scattering angle. For electron–positron/electron Coulomb scatttering, $\Delta\gamma(\alpha, \gamma, \gamma_1) = \frac{1}{2}(\gamma_1 - \gamma)(1-\alpha)$, and $d\sigma/d\alpha$ is the differential cross-section for: (i) 'Babha scattering' (e.g. BLP, equation 81.20) in the case of an electron scattering off a positron, or (ii) 'Möller scattering' (e.g. BLP, equation 81.10) in the case of an electron (positron) scattering off another electron (positron). The usual Coulomb divergence of equation (5.3) is eliminated by ignoring scatterings smaller than some angle θ_{min}, i.e. setting $\alpha_{max} = \cos(\theta_{min})$ where $\ln(1/\theta_{min}) \approx \ln\Lambda$, the Coulomb logarithm. For Babha scattering, $\alpha_{min} = -1$, while for Möller scattering (to avoid double counting), $\alpha_{min} = 0$.

When $\ln\Lambda \gtrsim 10$, ($\ln\Lambda$ is typically ~ 25 under conditions in AGN), $\langle\sigma\Delta\gamma\rangle_\alpha$ is well-approximated by

$$\langle \sigma\Delta\gamma\rangle_\alpha = \frac{3\sigma_T(\gamma_1-\gamma)}{64\varepsilon^2}\left[(4\ln\Lambda + 4\ln 2)\left(\frac{\varepsilon^2+p^2}{p^2}\right)^2 - 2\left(\frac{8\varepsilon^4-1}{p^2\varepsilon^2}\right) + \frac{12\varepsilon^4+1}{\varepsilon^4} - \frac{8p^2}{3}\frac{(\varepsilon^2+p^2)}{\varepsilon^4} + \frac{2p^4}{\varepsilon^4}\right] \tag{5.4}$$

in the case of Babha scattering, and

$$\langle\sigma\Delta\gamma\rangle_\alpha = \frac{3\sigma_T(\gamma_1-\gamma)}{32\varepsilon^2}\left[\left(\frac{\varepsilon^2+p^2}{p^2}\right)^2 (2\ln\Lambda + 1 - \ln 2) + 1/2 + 4\ln 2\right] \tag{5.5}$$

for Möller scattering. $\varepsilon = \{\frac{1}{2}[1+\gamma\gamma_1(1-\beta\beta_1\mu)]\}^{1/2}$ is the centre-of-mass energy of one of the incoming particles, and $p = \sqrt{\varepsilon^2-1}$. The remaining integrals are best evaluated numerically [although one more integration may be performed analytically by means of the variable substitution discussed in Haug (1988, equations 14 and 15)]. For $\tau \ll 1$, the integration over γ_1 may be done quickly by a Laguerre integration method. For $\tau \gtrsim 1$, however, care must be taken because of the large negative exponents that can arise if terms are not combined in the right order (e.g. as in equation 15 of Haug 1988).

In the limit of relativistic pairs scattering off a cool thermal-pair distribution ($\tau \ll 1$, $\gamma \gg 1$), it follows from equations (5.4, 5) that $\dot\gamma \approx -\frac{3}{2}\sigma_T c N_T \ln\Lambda$, or $1/\tau_{ee} = -\dot\gamma/\gamma \approx 4\pi c r_0^2 N_T \gamma^{-1}\ln\Lambda$, the result of Gould (1975). As noted there, the relevant value of Λ to be used is $\approx \gamma^{1/2} m_e c^2/h\omega_p$, where $\omega_p = (4\pi N_T e^2/m_e)^{1/2}$ is the plasma frequency. In calculations where τ is typically $\lesssim 0.1$ (e.g. Fabian *et al.* 1986), this approximation is adequate for energies $\gamma \gtrsim 10$. A better approximation may be constructed by interpolating between various limits given in Dermer (1985), Frankel, Hines & Dewar (1979), and Haug (1988):

$$\dot\gamma \approx -\frac{3\sigma_T c N_T \varepsilon (x-x_0)x^{1/2}}{2p(x^{3/2}+1)}\left(\frac{1}{1/(1+\tau)+2\tau}\right) A(\varepsilon), \tag{5.6}$$

where

$$A = 1/2 - \ln(\sqrt{2}\,\theta_{min}) + \left(\frac{\varepsilon-1}{2\varepsilon}\right)^2 (2\ln 2 + 1/4), \quad \theta_{min} = \frac{h\omega_p(\varepsilon_s^2+p_s^2)}{m_e c^2 \varepsilon_s p_s^2}, \tag{5.7}$$

$\varepsilon_s = [(\varepsilon+1)/2]^{1/2}$, $p_s = [(\varepsilon-1)/2]^{1/2}$, $x = (\varepsilon-1)/\tau$, $x_0 = 1+\tau/(1+\tau)$, $p = (\varepsilon^2-1)^{1/2}$, and ε is the energy of the *non*-thermal electron. This expression is good to within 10 per cent for $x \gtrsim 6x_0$, and to within ~ 25 per cent at $x \sim 2x_0$. The largest errors occur in the neighbourhood of $x = x_0$ since the fit near to $\dot\gamma = 0$ is not very accurate.

6 e–e BREMSSTRAHLUNG

Electron–electron (positron) bremsstrahlung is usually of minor importance in determining the photon- and electron-distribution functions. As a first step towards gauging what effects it may have, we consider two contributions: (i) radiation from a thermal electron–positron plasma; and (ii) emission by a non-thermal electron (positron) plasma interacting with a cold background electron–positron plasma. Contribution (i) has been dealt with extensively in the literature and good fits to cooling rates and emission spectra are available (see Haug 1975a,b, 1985a,b, 1987; Stepney & Guilbert 1983; Dermer 1984, 1986). To estimate contribution (ii), we have taken the cross-sections given in Haug (1975a, 1985a) (*cf.* also Alexanian 1968; Bayer, Fadin & Khose 1968) and numerically computed the cooling rates for an electron (positron) of energy γ as well as the corresponding emission spectra. Numerical tables (and the subroutines for generating them) are available. Note that the total radiated luminosity from non-thermal pairs (contribution ii) can be comparable to or exceed that from the thermal pairs alone (contribution i).

7 DISCUSSION

It has been realized over the past 10 years, that electron–positron pair plasmas are probably present in a variety of astronomical sources, and that they can be observed, both directly with γ-ray telescopes and indirectly through their influence on lower energy photons. Modelling of pair plasmas has been hindered by the complexity and strong energy-dependence of the cross-sections. A variety of approximations have been devised to deal with this, many of which have proven inadequate. In this paper, we have collected and developed a set of rate coefficients to deal with relevant elementary processes in the context of a kinetic approach to the problem (*cf.* Paper II). Kinetic approaches are most useful for dealing with environments where there are large ranges of photon frequency present and, consequently, large ratios of soft-photon to γ-ray densities. Their most serious deficiency is that radiative transfer can be easily handled in only the most primitive manner (but see Kusunose 1987).

The two most serious shortcomings of our work are the immediate assumption that the radiation and the pairs are isotropically distributed and the neglect of polarization effects. A small, tangled magnetic field will guarantee electron isotropy which will vindicate our assumption for electron–electron and electron–photon processes. However, under the conditions of small optical depth we encounter, pair production and photon-escape are not necessarily well simulated. Additional complications arise in expanding or highly magnetized sources where additional preferred directions are introduced into the problem. If the plasma contains a significant magnetic field then Faraday rotation can depolarize the radiation between scatterings at low frequencies (Blandford & Rees 1978). However, this will not occur at X- and γ-ray energies and, as the expected optical depths are small, large degrees of polarization might be measurable. This deserves further study.

ACKNOWLEDGMENTS

The authors are indebted to A. Fabian, G. Ghisellini, S. Phinney, R. Svensson and especially A. Zdziarski and P. Madau for extensive discussions of pair plasmas. The reader should be indebted to the referee for insisting upon a major shortening of the manuscript. Support under NSF grant AST 86-15325 and NASA grant NAGW1301 is gratefully acknowledged.

REFERENCES

Abramowitz, M. & Stegun, I. A., 1965. *Handbook of Mathematical Functions,* Dover, New York.
Aharonian, F. A. & Atoyan, A. M., 1981a. *Astrophys. Space Sci.,* **79,** 321.
Aharonian, F. A. & Atoyan, A. M., 1981c. *Phys. Lett.,* **99b,** 301.
Aharonian, F. A., Atoyan, A. M. & Nagapetyan, A. M., 1983a. *Astrophys.,* **19,** 323.
Aharonian, F. A., Kirillov-Ugryumov, V. G. & Kotov, Yu. D., 1983b. *Astrophys.,* **19,** 139.
Aharonian, F. A., Kirillov-Ugryumov, V. G. & Vardanian, V. V., 1985. *Astrophys. Space Sci.,* **115,** 201.
Alexanian, M., 1968. *Phys. Rev.,* **165,** 253.
Barbosa, D. P., 1982. *Astrophys. J.,* **254,** 301.
Baring, M., 1987a. *Mon. Not. R. astr. Soc.,* **228,** 681.
Baring, M., 1987b. *Mon. Not. R. astr. Soc.,* **228,** 695.
Bayer, V. N., Fadin, V. S. & Khose, V. A., 1968. *Soviet Phys.,* **26,** 1238.
Berestetski, V. B., Lifshitz, E. M. & Pitaevski, L. P., 1982. *Quantum Electrodynamics, Landau and Lifshitz Course of Theoretical Physics,* Vol. 4, 2nd edn, Pergamon Press, Oxford (BLP).
Bisnovatyi-Kogan, G. S., Zel'dovich, Ya. B. & Sunyaev, R. A., 1971. *Soviet Astr.,* **15,** 17.
Blandford, R. D. & Rees, M. J., 1978. In: *Pittsburgh Conference on BL Lac Objects,* p. 328, ed. Wolfe, A. M., University of Pittsburgh, Dept of Physics and Astronomy, Pittsburgh.
Blumenthal, G. R. & Gould, R. J., 1970. *Rev. Mod. Phys.,* **42,** 237.
Coppi, P. S., 1990. Caltech Theoretical Astrophysics Preprint GRP-209, *Mon. Not. R. astr. Soc.,* submitted.
Dermer, C. D., 1984. *Astrophys. J.,* **280,** 328.
Dermer, C. D., 1985. *Astrophys. J.,* **295,** 28.
Dermer, C. D., 1986. *Astrophys. J.,* **307,** 47.
Dermer, C. D. & Laing, E. P., 1989. *Astrophys. J.,* **339,** 512.
Fabian, A. C., Blandford, R. D., Guilbert, P. W., Phinney, E. S. & Cuellar, L., 1986. *Mon. Not. R. astr. Soc.,* **221,** 931.
Frankel, N. E., Hines, K. C. & Dewar, R. L., 1979. *Phys. Rev. A,* **20,** 2120.
Ghisellini, G., 1987a. *Mon. Not. R. astr. Soc.,* **224,** 1.
Ghisellini, G., 1987b. *PhD thesis,* ISSA, Trieste.
Gould, R. J., 1975. *Astrophys. J.,* **196,** 689.
Gould, R. J. & Schréder, G. P., 1967. *Phys. Rev.,* **155,** 1404.
Guilbert, P. W., 1981. *Mon. Not. R. astr. Soc.,* **197,** 451.
Guilbert, P. W. & Stepney, S., 1985. *Mon. Not. R. astr. Soc.,* **212,** 523.
Haug, E., 1975a. *Naturforsch,* **30a,** 1099.
Haug, E., 1975b. *Naturforsch,* **30a,** 1546.
Haug, E., 1985a. *Phys. Rev. D,* **31,** 2120.
Haug, E., 1985b. *Astr. Astrophys.,* **148,** 386.
Haug, E., 1987. *Astr. Astrophys.,* **178,** 292.
Haug, E., 1988. *Astr. Astrophys.,* **191,** 181.
Jones, F. C., 1965. *Phys. Rev.,* **137,** 1306.
Jones, F. C., 1968. *Phys. Rev.,* **167,** 1159.
Kusunose, M., 1987. *Astrophys. J.,* **321,** 186.
Kusunose, M. & Takahara, F., 1985. *Prog. Theor. Phys.,* **73,** 41.
Lamb, D. Q., 1988. *Am. Inst. Phys. Conf. Proc.,* **170,** 265.
Lightman, A. P. & Zdziarski, A. A., 1987. *Astrophys. J.,* **319,** 643.
Novikov, I. D. & Stern, B. E., 1986. In: *Structure and Evolution of Active Galactic Nuclei,* p. 149, eds Giuricin, G. *et al.,* Reidel, Dordrecht.
Protheroe, R. J., 1986. *Mon. Not. R. astr. Soc.,* **221,** 769.
Ramaty, R. & Mézsáros, P., 1981. *Astrophys. J.,* **250,** 384.
Ramaty, R. & Lingenfelter, R. E., 1987. *Am. Inst. Phys. Conf. Proc.,* **155,** 51.
Rybicki, G. & Lightman, A. P., 1979. *Radiative Processes in Astrophysics,* John Wiley & Sons, New York.
Stepney, S., 1983. *Mon. Not. R. astr. Soc.,* **202,** 467.
Stepney, S. & Guilbert, P. W., 1983. *Mon. Not. R. astr. Soc.,* **204,** 1269.
Stern, B. E., 1985. *Soviet Astr.,* **29,** 306.
Svensson, R., 1982. *Astrophys. J.,* **258,** 321.
Svensson, R., 1986. In: *Radiation Hydrodynamics in Stars and Compact Objects, IAU Coll. No. 89,* p. 325, eds Mihalas, D. & Winkler, K.-H. A., Springer-Verlag, Berlin.

Svensson, R., 1987. *Mon. Not. R. astr. Soc.*, **227,** 403.
Takahara, F., 1980. *Prog. Theor. Phys.*, **63,** 1551.
Zdziarksi, A. A., 1985. *Astrophys. J.*, **289,** 514.
Zdziarski, A. A., 1987. In: *Proc. 13th Texas Symp. Relativistic Astrophysics*, p. 553, ed. Ulmer, M., World Scientific Press, Singapore.
Zdziarski, A. A., 1988. *Astrophys. J.*, **335,** 786.
Zdziarski, A. A. & Lightman, A. P., 1987. In: *Proc. Workshop on Variability in Galactic and Extragalactic X-ray Sources*, p. 121, ed. Treves, A., Associazione per l'Avanzamento dell'Astronomia, Milan.

APPENDIX A

A1 Exact energy distribution of Compton scattered photons

Equation (20) of Jones (1968) appears incorrect and should read

$$\frac{d^4N}{dt\,d\alpha\,dz'\,d\eta}=\frac{cr_0^2\alpha}{2\gamma^4\alpha_1^2(\beta^2+\varepsilon^2+2\beta\varepsilon z')^{1/2}}\left[\frac{1+y_0^2+2y_0\delta\eta+\delta^2\eta^2}{(a+b\eta)}+\frac{(\alpha/\gamma)^2(1-y_0-\delta\eta)^2}{(1-\beta z')(a+b\eta)^2}\right]\frac{1}{(1-\eta^2)^{1/2}(1-\beta z')}. \tag{A1.1}$$

The subsequent equations (21) and (24–27) of Jones also appear incorrect. Integrating the corrected version of equation (20) over η, one obtains (changing notation slightly)

$$\begin{aligned}\frac{dP}{dz}(\omega;\omega',\gamma)=&\frac{3\sigma_T c\omega}{16\gamma^4\omega'^2R(\omega',\gamma)(\beta^2+\varepsilon^2+2\beta\varepsilon z)^{1/2}(1-\beta z)}\\&\times\Bigg[2y_0k-ak^2+(a^2-b^2)^{-1/2}[(1+y_0^2)-2ay_0k+a^2k^2]\\&+\frac{1}{k^2(1-\beta z)}\left\{k^2+k^2(a^2-b^2)^{-1/2}\frac{a(2b-a)}{(a-b)}+(a^2-b^2)^{-3/2}[a(1-y_0)^2+2kb^2(1-y_0)-ba^2k^2]\right\}\Bigg],\end{aligned} \tag{A1.2}$$

where $\varepsilon=\omega'/\gamma$, $k=\gamma/\omega$, $a=1-\beta z-(1-y_0)/k$, $b=\delta/k$, and

$$y_0=\frac{(\varepsilon+\beta z)(\rho+\varepsilon\rho-1+\beta z)}{\rho(\beta^2+\varepsilon^2+2\beta\varepsilon z)}, \tag{A1.3}$$

$$\delta=\frac{\beta(1-z^2)^{1/2}[\rho^2\beta^2+2\rho\varepsilon(1-\rho)(1-\beta z)-(\rho-1+\beta z)^2]^{1/2}}{\rho(\beta^2+\varepsilon^2+2\beta\varepsilon z)}, \tag{A1.4}$$

with $\rho=\omega/\omega'$. Somewhat time-consuming integrals needed to evaluate this are

$$\int_{-1}^{1}d\eta\frac{\eta^2}{(a+b\eta)(1-\eta^2)^{1/2}}=\pi\left[\frac{a^2}{b^2(a^2-b^2)^{1/2}}-\frac{a}{b^2}\right], \tag{A1.5}$$

and

$$\int_{-1}^{1}d\eta\frac{\eta^2}{(a+b\eta)^2(1-\eta^2)^{1/2}}=\pi\left[\frac{1}{b^2}+\frac{a}{b^2}\frac{(2b-a)}{(a-b)(a^2-b^2)^{1/2}}-\frac{a^2}{b(a^2-b^2)^{3/2}}\right]. \tag{A1.6}$$

The required energy distribution of Compton scattered photons is then given by

$$P(\omega;\omega',\gamma)=\int_{z_-}^{z_+}dz\frac{dP}{dz}$$

where

$$z_-=\max[-1,\beta^{-1}\{1-\rho[d+(d^2-1/\gamma^2)^{1/2}]\}], \tag{A1.7}$$

and

$$z_+=\min[+1,\beta^{-1}\{1-\rho[d-(d^2-1/\gamma^2)^{1/2}]\}], \tag{A1.8}$$

and $d=1+\varepsilon-\varepsilon\rho$. The minimum and maximum values of the scattered energy ω may be determined by using fig. 3 of Jones (1968) and following the accompanying discussion. Although the integral over z may be performed analytically, the resulting expression is even more complicated, and for some parameter regimes, difficult to evaluate numerically (see again Jones 1968). dP/dz, on the other hand, is relatively well-behaved and easily integrated numerically.

A2 Exact pair energy distribution for photon–photon pair production

Aharonian *et al.* (1983a) contains a derivation for the energy distribution $P(\gamma_\pm;\omega_1,\omega_2)$ of pairs produced in the annihilation of two photons with energy ω_1 and ω_2. Unfortunately the expressions given there appear to contain misprints. Also, we do not find the logarithmic divergence in the pair energy spectrum claimed for the case $\gamma_\pm=\omega_1=\omega_2$. An alternate expression may be derived by following the analysis of Svensson (1982) and making the minor modifications required for the case of photon–photon pair production. The result is

$$P(\gamma_\pm;\omega_1,\omega_2)=\frac{c}{\gamma_{\rm cm}\beta_{\rm cm}\gamma_{\rm c}\beta_{\rm c}R(\omega_1,\omega_2)}\int_{-1}^{+1}\frac{d\mu}{2}(1-\mu)\int_0^{2\pi}d\phi H(\gamma_{\rm cm}^2-1)\,H(1-|z|)\frac{d\sigma}{d\Omega_{\rm cm}}(\mu,\phi), \tag{A2.1}$$

where $\gamma_{\rm cm}^2=\frac{1}{2}\omega_1\omega_2(1-\mu)$, $\beta_{\rm cm}=(\gamma_{\rm cm}^2-1)^{1/2}/\gamma_{\rm cm}$, $\gamma_{\rm c}=(\omega_1+\omega_2)/2\gamma_{\rm cm}$, $\beta_{\rm c}=(\gamma_{\rm c}^2-1)^{1/2}/\gamma_{\rm c}$, $z=(\gamma_\pm-\gamma_{\rm cm}\gamma_{\rm c})/\gamma_{\rm cm}\gamma_{\rm c}\beta_{\rm c}\beta_{\rm cm}$, and

$$\frac{d\sigma}{d\Omega_{\rm cm}}(\mu,\phi)=\frac{3\sigma_{\rm T}\beta_{\rm cm}}{32\pi\gamma_{\rm cm}^2}\left\{-1+\frac{1}{2}(3-\beta_{\rm cm}^4)\left[\frac{1}{1-\beta_{\rm cm}x}+\frac{1}{1+\beta_{\rm cm}x}\right]-\frac{1}{2\gamma_{\rm cm}^4}\left[\frac{1}{(1-\beta_{\rm cm}x)^2}+\frac{1}{(1+\beta_{\rm cm}x)^2}\right]\right\}, \tag{A2.2}$$

where $x=yz+(1-y^2)^{1/2}(1-z^2)^{1/2}\cos\phi$, and $y=(\omega_1-\omega_2)/\beta_{\rm c}(\omega_1+\omega_2)$. $R(\omega_1,\omega_2)$ is the pair production rate (equation 4.3), and $H(x)$ is the Heaviside function.

This expression is easily evaluated numerically. We have verified (numerically) that it gives the correct first and second moments of the pair distribution (equations 4.8 and 4.9) and is properly normalized to unity [$\int P(\gamma_\pm;\omega_1\omega_2)\,d\gamma_\pm=1$]. We also obtain excellent agreement with the Aharonian *et al.* (1985) approximation in the region where the approximation is expected to work. The minimum and maximum possible pair energies for a given ω_1,ω_2 agree with the limits given in Aharonian *et al.* (1983a) (see their equation 22 and accompanying discussion).